

API Specification

Student OS Documentation — File 09

API Principles

Version APIs when appropriate, return consistent JSON, validate inputs, authenticate protected routes and return meaningful HTTP status codes.

Auth Endpoints

POST /auth/register

POST /auth/login

POST /auth/logout

GET /auth/me

Dashboard

GET /dashboard/summary

Timetable

GET /timetable

POST /timetable

PATCH /timetable/:id

DELETE /timetable/:id

Tasks

GET /tasks

POST /tasks

GET /tasks/:id

PATCH /tasks/:id

DELETE /tasks/:id

POST /tasks/:id/complete

Calendar

GET /calendar?from=&to=

Attendance

GET /attendance

GET /attendance/:subjectId

Exams

GET /exams

POST /exams

PATCH /exams/:id

DELETE /exams/:id

Notes

GET /notes

POST /notes

GET /notes/:id

PATCH /notes/:id

DELETE /notes/:id

Notifications

GET /notifications

POST /notifications/:id/read

GET /notification-preferences

PATCH /notification-preferences

Analytics

GET /analytics/attendance

GET /analytics/tasks

GET /analytics/study-hours

GET /analytics/exams

API Contract

Requests must validate schema. Responses should use consistent success/error shapes. Authorization must verify ownership/role. Pagination/filtering should be used for potentially large collections.

Errors

400 validation error; 401 unauthenticated; 403 unauthorized; 404 not found; 409 conflict; 429 rate limited; 500 unexpected server error.